REDUCING SUPPLIER NON-COMPLIANCE IN PROCUREMENT OPERATIONS

A DMAIC Case Study | LSS Green Belt Project

Dataset: Procurement KPI Analysis | Kaggle CC0 | 777 Purchase Orders | 2022-2023

PROJECT OVERVIEW

17.6% of 777 purchase orders violated compliance terms — putting $6.99M in contract value at risk. A structured DMAIC approach identified root causes and charted a path to $3.81M in projected recovery.

17.6%	2.43σ	$6.99M	$3.81M
NC Rate	**Sigma Level**	**Contract Value at Risk**	**Projected Save**
Of 777 POs reviewed	Baseline process capability	Total Negotiated value — NC orders	54.5% reduction in contract risk

APPROACH

- Chi-square CTQ validation
- Pareto analysis (76.6% in 2 suppliers)
- 5 Whys root cause
- Solution priority matrix
- Control plan + escalation triggers

DOCUMENT STRUCTURE

DEFINE › MEASURE › ANALYZE › IMPROVE › CONTROL › APEX › DISCLOSURE

PROBLEM STATEMENT

In 2022–2023, 17.6% of 777 purchase orders violated compliance terms, creating $6,986,051 in contract value at risk across 5 suppliers at Sigma Level 2.43 — below LSS-accepted benchmark of 3.0+.

Goal: Reduce NC from 17.6% → below 8% | Sigma 2.43 → 3.1+ | Recover ~$3.8M/year (54.5% of contract value at risk)

CTQ DECISION

✗ **Order Status**

Obvious. On every dashboard. REJECTED.

✓ **Compliance Rate**

Tested. Validated. SELECTED.

ELEMENT	DETAIL	VALUE	STATUS
Scope in	777 POs | 5 suppliers	2022–2023	Active
Scope out	Pricing | IT | Carriers SIPOC/VSM/Full COPQ — limited by dataset	Tier-2 suppliers	Excluded
Team	Helen Lam	Saint Leo Univ.	Academic

COMMENT

→ The obvious metric is the most dangerous choice — it has never been statistically validated.

→ Most Define failures are correct problem statements aimed at the wrong problem.

CTQ DECISION

Option 1: Order Status

p = 0.7612 ✗

All 5 suppliers: 24–30% non-delivery.
No pattern. No root cause.
3 hours of analysis = $0 finding.

Option 2: Compliance Rate

p = 0.0000000000 ✓

Delta_Logistics: 39.2% | Epsilon: 1.8%
22x gap between suppliers.
Clear pattern — actionable root cause.

METRIC	VALUE	CONTEXT
DPMO	176,319	Defects per million opportunities
Sigma Level	2.43	Target: 3.1+ | Below LSS-accepted benchmark of 3.0+
Contract Value at Risk	$6,986,051	Total value at risk from non-compliance

COMMENT

→ p = 0.76 vs p = 0.0000000000: same dataset, different CTQ, $6.99M difference in finding.

SUPPLIER	CASES	% TOTAL	PRIORITY
Delta_Logistics	67	48.9%	CRITICAL
Beta_Supplies	38	27.7%	HIGH
Gamma_Co	20	14.6%	MEDIUM
Alpha_Inc	9	6.6%	LOW
Epsilon_Group	3	2.2%	MINIMAL

Delta + Beta = 76.6% of ALL violations → Focus here first

COMMENT

→ Pareto without root cause = target without solution. Root cause without Pareto = solution without priority.

→ Root cause: structural absence, NOT incompetence. Wrong diagnosis = expensive, ineffective solution.

5 WHY – DELTA LOGISTICS

Why 1
CONTRACT TERMS VIOLATED — delivery + quality
[DATA: 67/171 orders = 39.2% NC]

Why 2
ACTUAL DELIVERY BELOW COMMITTED PERFORMANCE
[Delta lead time 11d avg vs Epsilon 4d]

Why 3
CAPACITY BELOW WHAT WAS COMMITTED AT SIGNING
[Why 3–5: structured inference — see Disclosure]

Why 4
CAPACITY ESTIMATES NEVER UPDATED AFTER SIGNING

Why 5
NO DYNAMIC CONTRACT REVIEW MECHANISM EXISTS

ROOT CAUSE: No dynamic contract review mechanism

SOLUTION	IMPACT	EFFORT	PRIORITY	TIMELINE
Monthly supplier compliance scorecard	★★★★★	★★	HIGH	Month 1
Delta 90-day improvement plan	★★★★★	★★★	HIGH	Month 1-3
Quarterly contract review mechanism	★★★★	★★★	HIGH	Month 2
Dual-source high-risk categories	★★★★	★★★★	MEDIUM	Month 3-6
Expand Epsilon_Group scope	★★★	★★	MEDIUM	Month 2-3

PROJECT ROI

METRIC	BEFORE	TARGET	IMPROVEMENT
Non-compliance Rate	17.6%	8.0%	**–54.5%**
Sigma Level	2.43	3.1+	**+27.6%**
Contract Value at Risk	$6,986,051	$3,175,478	**–$3,810,573/yr**

COMMENT

→ Highest-impact solution (monthly scorecard) requires least effort — data already exists.

→ The barrier was never capability — it was absence of a review cadence.

METRIC	TARGET	FREQUENCY	OWNER	TRIGGER
Compliance rate by supplier	>92%	Monthly	Procurement Mgr	Corrective plan
Defective units rate	<5%	Per delivery	QC Team	Reject + RCA
Sigma Level overall	>3.1	Quarterly	Process Owner	DMAIC review
Contract Value at Risk	−50% in 6mo	Monthly	Finance	Escalate to VP
Delta NC rate	<15% in 30d	Bi-weekly	Supplier Mgr	Escalate + dual-source
Contract review completion	100% quarterly	Quarterly	Procurement Lead	Halt new POs

OWNERSHIP

- Every metric needs a named owner.
- No owner = nobody's responsibility.

ESCALATION TRIGGERS

- Defined thresholds force action.
- Measurement without consequence = theater.

REVIEW CADENCE

- The root cause was absence of review.
- Control plan must institutionalize it.

COMMENT

→ A control plan without ownership is a wishlist. Without triggers, it is just measurement.

WHEN A MEASURE BECOMES A TARGET, IT CEASES TO BE A GOOD MEASURE.

What happens when Delta_Logistics knows exactly which metrics will be measured?

1. DEFINITION GAMING

Compliance Rate >92% Monthly

HOW IT'S GAMED:

Delta redefines 'compliant'— ±2 days tolerance.
Metric: GREEN.
Problem: UNCHANGED.

COUNTER-MEASURE:

Lock definition in contract before scorecard launches

→ Redefinition = contractual breach, not negotiation

2. BATCH SIZE GAMING

Defective Units <5% Per Delivery

HOW IT'S GAMED:

500 units, 24 defective = 4.8% (PASS)
250 units, 24 defective = 9.6% (FAIL)
Same defects. Different batch.

COUNTER-MEASURE:

Measure defect rate per unit, not per shipment

→ Batch size manipulation becomes irrelevant

3. ACTIVITY GAMING

Contract Review 100% Quarterly

HOW IT'S GAMED:

Attend. Sign. Walk out.
Change nothing.
Review: complete ✓
Outcome: zero.

COUNTER-MEASURE:

Add outcome metric: Delta NC rate 30d post-review

→ Links activity to measurable result

DATASET

- Kaggle CC0: Public Domain | Author: Shahriar Kabir
- 777 purchase orders | 2022–2024 | 5 anonymized suppliers

SCOPE & LIMITATIONS

- Root causes Why 3–5 are structured inferences — not validated through supplier interviews or confirmed implementation
- Why 1 (39.2% NC rate at Delta) is data-confirmed
- 11.2% missing delivery data acknowledged
- $6.99M = negotiated contract value of NC orders, not a P&L loss figure
- SIPOC / VSM / Full COPQ not included — limited by dataset structure

THE VALUE OF THIS WORK

- This project was conducted as a Green Belt candidate.
- The value of this work is not the conclusions. It is the methodology used to reach them — and the willingness to question that methodology before declaring the conclusions final.